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Clinical Trial

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Erlotinib versus standard chemotherapy as first-line treatment for European patients with advanced EGFR mutation-positive non-small-cell lung cancer (EURTAC): a multicentre, open-label, randomised phase 3 trial

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Abstract

Background: Erlotinib has been shown to improve progression-free survival compared with chemotherapy when given as first-line treatment for Asian patients with non-small-cell lung cancer (NSCLC) with activating EGFR mutations. **We aimed to assess the safety and efficacy of erlotinib compared with standard chemotherapy for first-line treatment of European patients with advanced EGFR-mutation positive NSCLC.**

Methods: **We undertook the open-label, randomised phase 3 EURTAC trial at 42 hospitals in France, Italy, and Spain. Eligible participants were adults (> 18 years) with NSCLC and EGFR mutations (exon 19 deletion or L858R mutation in exon 21) with no history of chemotherapy for metastatic disease (neoadjuvant or adjuvant chemotherapy ending $\geq$ 6 months before study entry was allowed). We randomly allocated participants (1:1) according to a computer-generated allocation schedule to receive oral erlotinib 150 mg per day or 3 week cycles of standard intravenous chemotherapy of cisplatin 75 mg/m² on day 1 plus docetaxel (75 mg/m² on day 1) or gemcitabine (1250 mg/m² on days 1 and 8). Carboplatin (AUC 6 with docetaxel 75 mg/m²) or AUC 5 with gemcitabine 1000 mg/m²) was allowed in patients unable to have cisplatin. Patients were stratified by EGFR mutation type and Eastern Cooperative Oncology Group performance status (0 vs 1 vs 2). The primary endpoint was progression-free survival (PFS) in the intention-to-treat population. We assessed safety in all patients who received study drug ($\geq$ 1 dose). This study is registered with ClinicalTrials.gov, number NCT00446225.**

Findings: **Between Feb 15, 2007, and Jan 4, 2011, 174 patients with EGFR mutations were enrolled. One patient received treatment before randomisation and was thus withdrawn from the study; of the remaining patients, 86 were randomly assigned to receive erlotinib and 87 to receive standard chemotherapy. The preplanned interim analysis showed that the study met its primary endpoint;**

enrolment was halted, and full evaluation of the results was recommended. At data cutoff (Jan 26, 2011), median PFS was 9.7 months (95% CI 8.4-12.3) in the erlotinib group, compared with 5.2 months (4.5-5.8) in the standard chemotherapy group (hazard ratio 0.37, 95% CI 0.25-0.54; $p < 0.0001$). Main grade 3 or 4 toxicities were rash (11 [13%] of 84 patients given erlotinib vs none of 82 patients in the chemotherapy group), neutropenia (none vs 18 [22%]), anaemia (one [1%] vs three [4%]), and increased amino-transferase concentrations (two [2%] vs 0). Five (6%) patients on erlotinib had treatment-related severe adverse events compared with 16 patients (20%) on chemotherapy. One patient in the erlotinib group and two in the standard chemotherapy group died from treatment-related causes.

Interpretation: Our findings strengthen the rationale for routine baseline tissue-based assessment of EGFR mutations in patients with NSCLC and for treatment of mutation-positive patients with EGFR tyrosine-kinase inhibitors.

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